

The Copperbelt University
MCS School of Medicine
Department of Basic Medical Sciences
3rd Year Preclinical (MBChB, BDS and BSc Clinical Medicine)
Academic year 2024/25

MBS 330 MEDICAL BIOCHEMISTRY – II

Rationale:

The Medical Biochemistry course is designed to provide the novice physician with a fundamental understanding of current concepts of human biochemistry and molecular biology. The aim of this course is to provide students with knowledge that will enable them appreciate how complex biomolecules are created and interconverted in our cells to provide energy, structure and function and carry out basic biochemical techniques. Collectively understanding biochemistry will also benefit the students as an essential building block to better understanding of other basic medical sciences courses (physiology, cell and molecular biology, pathology, microbiology, immunology, toxicology and pharmacology) in their education. This biochemistry course with metabolic approach, will focus on biochemical pathways involved in intermediary metabolism, and also look at the pathways of DNA replication, transcription and processing, protein synthesis/ processing/ targeting/ degradation to understand the concepts, mechanisms and regulation of gene expression and control. Students are also trained in a variety of basic biochemistry-based laboratory skills.

Pre-requisite course: MBS230 - Medical Biochemistry-I

Learning Objectives:

After learning this course the student should be able to:

1. Describe the concepts, principles of metabolic pathways and their significances for medicine.
2. Describe the sub cellular localization of metabolic pathways and understand the coordinated inter relationships and their regulations.
3. Differentiate the anabolic and catabolic pathways and their physiological importance.
4. Describe the reactions of various metabolic pathways and their correlations to health and disease
5. Estimate energy requirements of our body, energy yield of various metabolites and their thermodynamic characteristics.
6. Appreciate how regulations of biochemical pathways lead to normal integrated metabolism and cellular homeostasis.
7. Extrapolate how abrogation of normal integrated metabolism can result in various

disease states.

8. Describe the processes of DNA replication, DNA damages and repairs, RNA synthesis and processing, and protein synthesis/ processing/ targeting/ degradation and their importance in health and disease
9. Appreciate the mechanism of regulation of gene expression and control in prokaryotes and eukaryotes to understand the mechanism of action of therapeutics drugs.
10. Understand the biochemical basis of human diseases, including metabolic, genetic and inherited conditions.
11. Define genetic engineering and its applications in medicine and to explain the basic principles of genetic engineering
12. Describe the various genetic engineering techniques and discuss the applications of genetic engineering in medicine

Learning outcomes:

1. Knowledgeable on chemical reactions of life and integration of the various aspects of metabolism, and their regulatory pathways
2. Describe the biochemical basis of inherited disorders and their associated sequelae.
3. Capacity to explain the normal biochemical functioning of human body and alterations in health and diseases
4. Ability to predict and link the metabolic pathways, which could be defective under various diseases and conditions.
5. Ability to identify errors in metabolism and explain how specific disease states can arise due to these errors.
6. Ability to identify the biochemical markers for various metabolic conditions.
7. Ability to connect the biochemical markers to the health and disease in human.
8. Capacity to understand the scenarios/ case files related to metabolic conditions and explain the biochemical basis for their manifestations.
9. Ability to explain molecular mechanisms of gene expression and regulation, the principles of genetic engineering and their application in medicine
10. Ability to analyze and interpret a given data; the ability to make clinical diagnosis and communicate complex biochemical concepts effectively.
11. Apply the biochemical principles to novel clinical scenarios and research questions.

Lecture schedule for Term: I

WEEK	TOPICS	LECTURER
1	OVERVIEW OF METABOLISM ✓ Introduction and major functions ✓ Catabolic and Anabolic reactions ✓ Differences between Catabolism and Anabolism ✓ Principles of Metabolic pathways ✓ Stages of Metabolism ✓ Medical importance of Metabolism ✓ Types of chemical reactions based on free energy change ✓ Mechanism of energy Coupling ✓ Role of ATP in chemical reactions of our body	Dr Balakrishnan S
2	ELECTRON TRANSPORT CHAIN AND OXIDATIVE PHOSPHORYLATION ✓ Introduction to Biological Oxidation ✓ Nature of Biological Oxidation ✓ Structure of ETC ✓ Functioning of ETC ✓ Mitochondrial shuttle systems ✓ Oxidative phosphorylation ✓ Substrate level phosphorylation ✓ Chemiosmotic theory ✓ Coupling of ETC and Oxidative Phosphorylation ✓ Regulation of ETC and Oxidative Phosphorylation ✓ Inhibitors of ETC (a case file / scenario with clinical correlations) ✓ Deficiency disorders of ETC and oxidative phosphorylation – OXPHOS diseases (a case file / scenario with clinical correlations)	Dr Balakrishnan S
3	GLYCOLYSIS ✓ BioMedical importance of Glycolysis ✓ Reactions and regulation of the Glycolytic pathway ✓ Importance of glycolysis for normal red blood cells ✓ Consequences of deficiencies in glycolytic enzymes ✓ Disorder of Pyruvate kinase deficiency (a case file / scenario with clinical correlations) ✓ Energy calculations in the glycolytic pathway ✓ Fate of pyruvate ✓ Metabolic Reprogramming in cancer cells & Warburg effect (a case file / scenario with clinical correlations)	Dr Balakrishnan S

	Metabolism of lactose & galactose: ✓ Lactose intolerance, galactosemia (a case file / scenario with clinical correlations) Metabolism of sucrose & fructose: ✓ Fructose intolerance & Fructosemia (a case file / scenario with clinical correlations) ✓ Polyol pathway	
4	PENTOSE PHOSPHATE PATHWAY (PPP) ✓ BioMedical importance of pentose phosphate pathway ✓ Steps, reactions, and regulation of pentose phosphate pathway ✓ Relationship between PPP and glycolysis. ✓ Disorder of G6PD deficiency (a case file / scenario with clinical correlations) ✓ G6PD deficiency and malaria (a case file / scenario with clinical correlations) ✓ G6PD deficiency and essential hypertension (a case file / scenario with clinical correlations) ✓ G6PD deficiency and Favism (a case file / scenario with clinical correlations)	Dr Balakrishnan S
5	URONIC ACID PATHWAY ✓ BioMedical importance of Uronic acid pathway ✓ Steps and reactions of Uronic acid pathway ✓ Clinical importance of Uronic acid pathway ✓ Essential pentosuria (a case file / scenario with clinical correlations) GLUCONEOGENESIS ✓ BioMedical importance of Gluconeogenesis ✓ Pathway of gluconeogenesis and its regulation ✓ Differences and similarities between gluconeogenesis and glycolysis ✓ Reciprocal control of glycolysis and gluconeogenesis ✓ Cori cycle ✓ Regulation of Blood Sugar – hormonal regulation ✓ Glycosuria and Hyperglycemic conditions (a case file / scenario with clinical correlations)	Dr Balakrishnan S
7	EFFECT ALCOHOL ON GLUCOSE METABOLISM ✓ Pathways of Alcohol metabolism ✓ Gluconeogenesis Vs alcohol metabolism ✓ Effect alcohol on liver metabolism (a case file /	Dr Balakrishnan S

	scenario with clinical correlations) TRICARBOXYLIC ACID CYCLE (Krebs cycle) ✓ BioMedical importance ✓ Steps and reactions of the TCA ✓ Regulation of the TCA cycle ✓ Pyruvate dehydrogenase deficiency ✓ Energetics of TCA ✓ Amphibolic Nature of the TCA cycle ✓ Anaplerotic Nature of the TCA cycle ✓ Integration of Metabolism ✓ Link between the TCA cycle and the electron transport chain. ✓ Relation to Glyoxylate cycle	
8	GLYCOGEN METABOLISM ✓ BioMedical importance of glycogen metabolism ✓ Reactions and regulation of Glycogenesis ✓ Reactions and regulation of Glycogenolysis ✓ Hormonal control of glycogen metabolism ✓ Glycogen storage diseases (a case file / scenario with clinical correlations)	Dr Balakrishnan S
9	End of Term Tests (CA)	Dr Balakrishnan S
10	End of Term Tests (CA)	Dr Balakrishnan S

Lecture schedule for Term: II – Dr Kwangu M

Lecture schedule for Term: III

WEEK	LECTURE NO.	TOPICS	LECTURER
3	1	NUCLEOTIDE METABOLISM: Biological importance of nucleotide metabolism PURINE NUCLEOTIDE METABOLISM: De novo synthesis of Purine Nucleotides. Salvage Pathways of Purine Nucleotides Synthesis Regulation of Purine Nucleotides Synthesis.	Dr S. Balakrishnan
	2	PURINE NUCLEOTIDE METABOLISM (Cont.): Degradation of Purine Nucleotides. Disorders of Purine metabolism. PYRIMIDINE NUCLEOTIDE METABOLISM:	Dr S. Balakrishnan

		De novo synthesis of pyrimidines.	
	3	PYRIMIDINE NUCLEOTIDE METABOLISM (Cont.): Salvage Pathways of pyrimidine nucleotides synthesis Regulation of pyrimidines synthesis. Synthesis of deoxyribonucleotides Catabolism of pyrimidines. Disorders of pyrimidine metabolism.	Dr S. Balakrishnan
	4	CENTRAL DOGMA OF A CELL: DNA REPLICATION: Overview of DNA replication Salient features of DNA replication Enzymes involved in DNA replication Steps and mechanism of DNA replication	Dr S. Balakrishnan
4	5	DNA REPLICATION (Cont.): Telomeres problem in eukaryotes Differences between replication in prokaryotes and eukaryotes Inhibitors of DNA replication	Dr S. Balakrishnan
	6	DNA DAMAGES AND DNA REPAIR MECHANISM: Types of DNA damage Diverse DNA repair systems: Direct repair, e.g. pyrimidine dimers Excision repairs Mismatch excision repair Double-strand break repair and recombination Diseases related to DNA repair defects	Dr S. Balakrishnan
5	7	TRANSCRIPTION AND RNA PROCESSING: Overview of transcription Comparison of prokaryotic and eukaryotic RNA polymerases Transcription promoters in prokaryotes and eukaryotes	Dr S. Balakrishnan

		Steps and mechanism of transcription Rho-independent and rho-dependent termination	
	8	TRANSCRIPTION AND RNA PROCESSING (Cont.): Inhibitors of DNA-Dependent RNA Polymerases Differences between prokaryotic and eukaryotic transcription Post-transcriptional processing Reverse transcription Synthesis of cDNA	Dr S. Balakrishnan
6	9	TRANSLATION (Synthesis of Protein) Genetic code and Wobble Hypothesis Overview of translation Protein synthesis machinery Amino Acid Activation Steps of translation Inhibitors of protein synthesis	Dr S. Balakrishnan
	10	POSTTRANSLATIONAL MODIFICATIONS: Protein folding and subunit assembly Protein targeting Similarities of prokaryotic and eukaryotic translation Difference between prokaryotic and eukaryotic translation	Dr S. Balakrishnan
7	11	CONTROL OF GENE EXPRESSION IN PROKARYOTES: Positive regulation and Negative regulation Operon models: Repression – Lac operon Derepression – Lac operon Attenuation – Trp operon	Dr S. Balakrishnan

	12	CONTROL OF GENE EXPRESSION IN EUKARYOTES: RNA processing Stability of m-RNA Gene amplification Gene rearrangement Gene regulation by histones and nonhistone proteins Class switching Binding of regulatory proteins to DNA Role of Enhancers, Silencers, Locus control regions (LCRs) and Insulators RNA interference	Dr S. Balakrishnan
8	13	GENETIC RECOMBINATION & MEDICINE: Introduction to Recombinant DNA (rDNA) and its application in medicine Basic steps in rDNA technology Transgenic animals and its application in medicine Genome library and cDNA library	Dr S. Balakrishnan
	14	GENETIC TECHNIQUES & ITS USES IN MEDICINE: DNA probes Blot transfer techniques. Polymerase chain reaction (PCR), Restriction fragment length polymorphism and its applications. Fluorescent in-situ hybridization (FISH). DNA sequencing. DNA fingerprinting and DNA footprinting. Karyogram and karyotyping.	Dr S. Balakrishnan
9	--	End of Term Tests (CA)	--
10	--	End of Term Tests (CA)	--

Teaching and Learning Methods:

- Lectures - 2 hours per week
- Practical - 3 hours per fortnight

Online Platform for Interaction:

- <https://classroom.google.com/c/NzYyNTg2OTQxNTkw?cjc=o5yqqye3>

Assessments:

- Continuous assessment - **40%**
 - End of term tests and Laboratory activities
- Final examination - **60%**

PRESCRIBED TEXTBOOKS

1. Robert K. Murray, David A. Bender, Kathleen M. Botham, Peter J. Kennelly, Victor W. Rodwell, P. Anthony Weil. "Harper's illustrated Biochemistry." Twenty-ninth Edition, Mc Graw Hill Lange, international Ed, 2012
2. Baynes J.W and Dominiczack M.H. Medical Biochemistry. 3rd Edition 2009

RECOMMENDED TEXTBOOKS

1. "Basic Medical Biochemistry- A Clinical Approach", by Dawn B. Marks, Allan D. Marks and Colleen M. Smith, Williams and Wilkins, Baltimore, 1996. ISBN 0-683-05595-X
2. "Principles of Biochemistry", 3rd Edition, by Albert L. Lehninger, David L. Nelson and Michael M. Cox, Worth Publishers, New York, 2000. ISBN:1-57259-153-6.
3. "Textbook of Biochemistry with Clinical Correlations", 4th Edition, Thomas M. Devlin, Ed., Wiley-Liss, Inc., New York, 1997. ISBN: 0-471-15451-2.
4. "Biochemistry", by Donald Voet and Judith Voet, Wiley-Liss, Inc., New York, 1995 ISBN: 0-471-58651-X.
5. "Principles of Biochemistry", by Geoffrey Zubay, William W. Parson and Dennis E. Vance, William C. Brown Publishers, Dubuque, IA, 1995. ISBN:0-697-14275-2.
6. "Biochemistry", Fourth Edition, by Lubert Stryer, W.H. Freeman and Co., New York, 1995. ISBN:0-7167-2009-4.
7. "Biochemistry", 2nd Edition, by Pamela C. Champe and Richard A. Harvey, J.B. Lippincott Co., Philadelphia, PA, 1997. ISBN: 0-397-51091-8.
8. Textbook of Medical Biochemistry, 8th Edition by M. N. Chatterjea, JP Medical Ltd, 2011